

Context and Understanding of Knowledge

Prompt: *In the acquisition of knowledge, can we only understand something to the extent that we understand its context?*

When I was eight, I became obsessed with the periodic table. My father couldn't understand why. He assumed I liked the letters, or maybe I had watched some flashy YouTube videos about it. It wasn't until I told him "without the periodic table, you and I don't exist" that he got it. I saw the periodic table as the building blocks of reality itself. He saw a child who was off or exceptional. Same object, entirely different inferences. The gap wasn't intelligence. It was context.

As a neurodivergent person, I've always operated with different contexts: sensory, perceptual, cognitive. I understand others better when I can see the context they're coming from: why people visit crowded temples, practice certain superstitions, what their fears mean to them. Context is what makes you go "oh, this makes sense now." It grounds your emotions: disappointment, anger, excitement, or joy when responding to anything. Without it, people feel incomprehensible, their actions arbitrary.

I think the same principle applies to knowledge itself. Whether we're trying to understand a person's behavior or a scientific finding, context turns raw observation into meaning. Understanding why someone practices a particular ritual or what a physics equation describes both require grasping the surrounding circumstances and purposes.

This pattern made me ask: can we only understand something to the extent that we understand its context? My answer is yes, largely, with one refinement. Knowledge is always situated in context, but understanding requires metacognition of that context: recognizing which contextual factors matter and why. The challenge is calibrating relevant context, not capturing it all. I use *context* to mean the aims, assumptions, circumstances, and constraints that make a claim intelligible, and understanding to mean an explanatory grasp that can be used to predict, intervene, generalize, or transfer insight.

I will argue that inquiry moves through a cycle: we first isolate context by constraining what varies—defining control variables in science experiments, fixing assumptions in mathematics, or specifying training data in AI. This isolation allows us to see patterns and relationships clearly. We then apply this understanding in real-world contexts to test whether it holds, for example checking external validity in medicine, testing AI systems across different use cases or environments, or seeing if mathematical models fit new data. Finally, we refine our understanding based on what we learn when context expands and situate knowledge in its appropriate context. I develop this through two Areas of Knowledge: the Natural Sciences, focusing on medicine, and Mathematics, examining how context operates in both abstract proofs

and real-world applications. Along the way, I address counterclaims: that mathematical truth is universal, that science discovers laws by removing context, and that demanding context leads to paralysis or relativism.

Natural Sciences: when signals need stories

In medicine, understanding emerges from correlating across layers of context. Clinical symptoms, diagnostic test results, and environmental factors mean nothing in isolation. A fever tells you little by itself. An elevated white blood cell count is just a number. But fever plus elevated WBC plus recent travel to a malaria-endemic region starts to mean something. Medical practice is the continuous work of correlating clinical presentation with diagnostic markers and situating both within the patient's specific context: their history, their environment, what is currently circulating in the population.

Medical knowledge, such as how symptoms correlate with diseases or how diagnostic numbers indicate functional deficiencies, only becomes understanding when context defines the connections. When I had COVID, my test results showed several abnormalities, but we initially missed the diagnosis because the presenting symptom—severe throat pain—didn't match the expected profile from earlier strains. I had context—knowledge of how COVID typically presented—but it was outdated. Understanding required knowing the new strain behaved differently. My understanding of the test results was limited by the extent of my contextual knowledge: I could only interpret the data correctly once I understood the strain-specific context. The same diagnostic pattern meant different things depending on which variant framework I used to interpret it. This suggests that in medicine, our understanding of what data means is constrained by how complete and current our contextual knowledge is. Partial or outdated context doesn't just limit understanding, it can produce misunderstanding.

Yet clinicians cannot wait for complete context before acting. Antibiotics are started before blood cultures return, based on limited context: fever, suspected infection site, local resistance patterns. Elevated cholesterol gets treated even without symptoms because the numbers plus risk factors provide enough context to justify intervention. This isn't reckless action; it is understanding that operates with deliberately limited context, ready to adjust as more information arrives.

Counterclaim: progress by isolating context

If context is so decisive, how does science discover general truths? The Natural Sciences advance by methodically **isolating** context. Controls, randomized trials, and standardized protocols hold background factors constant so patterns can emerge. This is an epistemological necessity. The researcher using controlled conditions follows the same logic as the clinician treating before complete data arrive: isolate what is decision-relevant now, discover or act, then test whether the insight transfers across broader contexts.

We constrain complexity to see structure, then expand context to test whether that structure holds beyond the lab. Discovery passes through isolated simplicity; application passes through restored complexity. So while understanding is context-bound, science progresses by strategically stripping context to reveal relationships, then re-embedding those relationships into more complex, real-world contexts.

Mathematics: abstract proofs and real-world applications

Pure mathematics appears to challenge the contextual thesis. Anyone can understand the Pythagorean theorem. Carpenters use it to check if corners are square; sailors use it for navigation; both may do so without formal mathematical training. Their understanding seems independent of context.

But their understanding is limited to specific application contexts. The carpenter knows it works for right-angled corners; the sailor knows it works for navigation triangles. A mathematics student learning systematically—polygons to triangles to right angles to the theorem itself—gains contextual scaffolding that enables transfer across many contexts. This deeper understanding comes from the layered context of formal mathematics.

Even Pythagoras himself operated within the context of Euclidean geometry: flat space where parallel lines never meet. He didn't know curved spaces existed. The theorem's truth depends on which geometric context you are in. In non-Euclidean geometries, the familiar relationship between the sides of a right triangle no longer holds in the same way. Understanding the theorem properly means knowing *where* it holds and *why*. Pure mathematics doesn't eliminate context; it **fixes** context explicitly through axioms and definitions, then builds understanding within those constraints.

Applied mathematics: when patterns meet the world

AI is applied mathematics, finding patterns in data to map inputs to outputs. This works brilliantly until context shifts. Image classifiers confidently misidentify adversarial examples—slightly modified images humans easily recognize. A stop sign with carefully placed stickers can be classified as “speed limit” with 99% confidence. The AI has learned correlations in its training context but has no understanding of what “stop sign” means beyond pixel patterns.

Large models produce fluent but factually wrong answers, hallucinate citations, and invent details because they are extending patterns beyond the contexts in which those patterns were learned. These errors happen because AI lacks metacognition about its own context. It doesn't know what it doesn't know. It can't recognize when it is operating outside the domains where its learned relationships are valid.

Current AI research increasingly focuses on context-aware systems and out-of-distribution detection—ways of making models cognizant of their context boundaries. This demonstrates that in applied mathematics, understanding is limited by whether the model’s learned context matches the world’s actual context. We understand the patterns only to the extent that we understand the contexts that generated them and the contexts where they apply.

Ethics: whose context counts?

Whose context counts is not just about adding stakeholder perspectives. It is about power. Economic, political, and social structures shape what gets recognized as relevant context. Those with power define context in ways that serve their interests.

Medical research historically used male subjects as the neutral default, making women’s physiological context invisible. Dosing guidelines, symptom profiles, and drug efficacy studies reflected male biology. This was not neutral science; it was power shaping what counted as relevant medical context. Many women’s heart attack symptoms, for instance, were under-recognized because the standard context was male.

In AI, training data reflects who has power to collect, label, and define ground truth. Facial recognition systems trained primarily on white faces fail on other racial contexts because those contexts were marginalized in data collection. The resulting systems encode whose context mattered to the people building them.

Recognizing relevant context is therefore epistemic, moral, and political. Methods alone do not solve this. We must ask who sets the methods, who defines relevant, and who benefits from context being framed this way. Understanding is constrained not just by which context we grasp, but by which contexts power structures make visible or invisible. Context-blind claims to universal knowledge often mean context was stripped in ways that served particular interests.

Conclusion

So, can we only understand something to the extent that we understand its context? Yes, with one crucial refinement: understanding is not about capturing *all* context, but about calibrating *relevant* context. The isolate–apply–refine cycle shows how this works. In medicine, I couldn’t understand my test results until I grasped the strain-specific context; outdated context produced misunderstanding. In pure mathematics, understanding depends on the explicitly fixed contextual foundations—axioms and geometric assumptions—that make theorems true. In applied mathematics, AI systems fail when their learned context does not match the world’s actual context, revealing that pattern recognition without contextual grounding is not genuine understanding. And across all domains, whose context counts is shaped by power structures that make some contexts visible and others invisible.

Ultimately, understanding context well requires more than documentation and procedures. It demands values and capacities. Values like fairness, inclusion, equality, justice, and integrity help determine whose contexts matter and why. Without these values, we risk systematically ignoring contexts that do not serve power. Individual capacities—critical thinking, metacognition, and self-awareness—enable us to recognize our own contextual blind spots and question what we have taken for granted. To the extent that we cultivate these values and capacities, our understanding of context deepens, and with it, our understanding of knowledge itself.

Note: *This essay was written for the IB Theory of Knowledge course as part of the IB Diploma Core. The task required responding to a prescribed prompt by developing a sustained argument, using examples from multiple areas of knowledge, and addressing counterclaims.*